

H15 Zhussupbekova Zere (T-2401

Instance $N_{srv} = 3$ capacities $C = (3, 5, 6)$ total demand $D = 6$

5.1 at $z = (0, 0)$

Pinned softmax $w_j = e^{z_j} / (1 + \sum_{k=0,1} e^{z_k})$ for $j = 0, 1$; $w_2 = 1 / (1 + \sum_{k=0,1} e^{z_k})$
 at $z = (0, 0)$: $e^0 = 1$ for both free logits so denom = $1 + 1 + 1 = 3$

$$W = \left(\frac{1}{3}; \frac{1}{3}; \frac{1}{3}\right)$$

Utilization $u_j = D w_j / C_j$

$$u = \left(\frac{6 \cdot \frac{1}{3}}{3}, \frac{6 \cdot \frac{1}{3}}{5}, \frac{6 \cdot \frac{1}{3}}{6}\right) = \left(\frac{2}{3}, \frac{2}{5}, \frac{1}{3}\right) = (0.6667, 0.400, 0.333)$$

$$J = \frac{1}{N} \sum_j u_j^2 = \frac{1}{3} (0.6667^2 + 0.4^2 + 0.333^2) = \frac{1}{3} (0.444 + 0.1600 + 0.111) = \frac{0.7156}{3} = 0.2385$$

Gradient by chain rule First: $s_{j1} = \frac{dJ}{dw_j} = \frac{2D^2}{N} \cdot w_j / C_j^2$ here $\frac{2D^2}{N} = 24$

$$s_0 = \frac{24 \cdot \frac{1}{3}}{9} = \frac{8}{9} = 0.889 \quad s_1 = \frac{24 \cdot \frac{1}{3}}{25} = 0.32 \quad s_2 = \frac{24 \cdot \frac{1}{3}}{36} = \frac{8}{36} = 0.222$$

$$\sum_j w_j s_j = \frac{1}{3} (0.889 + 0.32 + 0.222) = \frac{1.431}{3} = 0.477$$

Then $\frac{dJ}{dz_k} = w_k (s_k - \sum_j w_j s_j)$ for 3 coordinates $k = 0, 1$

$$\frac{dJ}{dz_0} = \frac{1}{3} (0.889 - 0.477) = \frac{0.4119}{3} = 0.1373$$

$$\frac{dJ}{dz_1} = \frac{1}{3} (0.32 - 0.477) = \frac{-0.157}{3} = -0.0523$$

$$\nabla_z J(0, 0) = (0.1373; -0.0523)$$

5.2 Closed-form minimizer and comparison

$$w_j^* = \frac{C_j^2}{\sum_k C_k^2} = \frac{C_j^2}{9 + 25 + 36} = \frac{C_j^2}{70} \Rightarrow w^* = \left(\frac{9}{70}, \frac{25}{70}, \frac{36}{70}\right) = (0.1286, 0.3571, 0.5143)$$

$$J^* = \frac{D^2}{N \sum_k C_k^2} = \frac{36}{3 \cdot 70} = \frac{36}{210} = 0.1714$$

Check the gradient vanishes there $z_j^* = \ln(w_j^* / w_2^*)$ for $j = 0, 1$

$$z_0^* = \ln\left(\frac{9}{36}\right) = \ln(0.25) \approx -1.3863 \quad z_1^* = \ln\left(\frac{25}{36}\right) = \ln(0.6944) \approx -0.3646$$

substituting z^* back into softmax-pinned reproduces w^* exactly and repeating the gradient computation at the previous subsection at w^* gives $s_j^* \propto w_j^*/C_j^2 = C_j^2 / (\sum_k C_k^2) / C_j^2 = 1 / \sum_k C_k^2$ so $s_k^* = \sum_j w_j^* s_j^* = 0$ for every k and therefore $\nabla_z J(z^*) = 0$ exactly

capacity proportional split

$$w_{\text{prop}} = \left(\frac{3}{14}; \frac{5}{14}; \frac{6}{14} \right) = (0.2143; 0.3571, 0.4286)$$

$$u_{\text{prop},j} = \frac{D w_{\text{prop},j}}{C_j} = \frac{D}{\sum_k C_k} = \frac{6}{14} = 0.4286 \quad \forall j$$

$$J_{\text{prop}} = \frac{1}{3} \cdot 3 \cdot (0.4286)^2 = (0.4286)^2 = 0.1837$$

$$J^* = 0.1714 < J_{\text{prop}} = 0.1837 \rightarrow \text{equal utilization is not the minimizer}$$

Why? minimizing $\sum_j u_j^2$ subject to the linear constraint $\sum_j C_j u_j = D$ is by Lagrange/Cauchy-Schwarz solved by $u_j \propto C_j$ - i.e. $w_j \propto C_j^2$ not $w_j \propto C_j$ a unit of extra share sent to large capacity server raises its utilization by less marginal cost to $\sum u_j^2$ than sending that same unit to a small server - so the optimum pushes utilization up on big servers and down on small ones past the point of equal utilization

Zhusup Bakore Zere 